Coding & Solution

Date	7 July 2026
Team ID	**XXXXX**
Project Name	Smart Lender
Maximum Marks	5 Marks

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1. Solution Summary Table

Field	Details
Repository Link / URL	https://github.com/Nikhil-tech664/Smart-Lender-Loan-Eligibility-Prediction
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	• Loan Eligibility Prediction • Data Preprocessing & Feature Engineering (SMOTE) • Machine Learning Model Training & Comparison • XGBoost Prediction Engine (rdf.pkl) • Flask Web Application • Responsive Glassmorphic User Interface • Client & Server-side Input Validation • Indian Currency Formatting (Lakhs & Rupees) • Underwriting Rule Vetoes (Credit & DTI) • Cloud Deployment using Render
Pending / Incomplete Features	• User Authentication (Future Enhancement) • Database Integration (Future Enhancement) • Admin Dashboard (Future Enhancement) • Applicant History Management (Future Enhancement)
Setup / Run Instructions	1. Clone the repository. 2. Create and activate a Python virtual environment. 3. Install dependencies using pip install -r requirements.txt. 4. Ensure the dataset and trained model files are available. 5. Run the application using python app.py. 6. Open http://127.0.0.1:5000 in a web browser.

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2. Code Quality Checklist Table

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functional modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

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3. Additional Notes / Comments

- **The application follows a modular project structure with separate folders for templates, static assets, datasets, machine learning models, and utility modules.**
- **Multiple Machine Learning algorithms (Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and XGBoost) were trained and evaluated to identify the best-performing model.**
- **XGBoost was selected as the final prediction model based on overall evaluation metrics (94.7% training accuracy and 81.1% testing accuracy).**
- **The application provides real-time loan eligibility prediction through a simple and responsive Flask web interface.**
- **The project is deployed on Render and maintained using GitHub for version control and collaboration.**
- **The architecture has been designed to support future enhancements such as authentication, database integration, and additional financial services.**